# Generating Best Frame for a Video Using GenAI

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***Abstract*—This project introduces the most comprehensive AI pipeline yet to integrate audio extraction, speech-to-text transcription, text summarization, and image generation into a no-fuss framework. The current venture fills an important gap that had been left in this domain of multimedia content analysis and visualization, where each described activity is usually performed separately. This system put them all together and finally presented an innovative end-to-end solution for content automated processing and visualization. The pipeline leverages the state-of-the-art models of machine learning, including Whisper by OpenAI for transcription with high accuracy and robustness, transformer-based models like T5, GPT 3.5, BART and PEGASUS for coherence and conciseness, and DALL·E to generate compelling and contextually relevant images based on summarized content. It was designed to take diverse multimedia inputs comprising podcasts, lectures, and interviews and went through vigorous evaluations of performance for metrics comprising transcription accuracy, coherence of summarization, and relevance and quality of images generated. These are therefore effective means by which such complicated audio-visual data gets summed up into summarized information with relevant illustrative visuals, thereby rendering it quite useful in many content creation applications, educational tools, and multimedia summaries. For example, educators can have the system process long lectures into concise, illustrated notes, while content creators can create visual summaries for blog posts and social media. Beyond the immediate applications, this project contributes to the study of artificial intelligence by demonstrating what can be achieved with integrated AI systems. It shows the synergy achieved by combining state-of-the-art models into one cohesive workflow. This also opens the way to several further innovations: real-time processing, multilingual, and domain-specific adaptations, among others, further expand the applicability and impact of the pipeline across industries.**

***Index Terms*—Integrated AI Systems, Speech-to-Text Transcription, Text Summarization, Image Generation**

## I. Introduction and Statement of the Problem

Exponential growth in the amount of multimedia data, including audio, video, text, and images, has imposed serious challenges on modern-day digital era requirements for efficient systems that could process, analyze, and mine useful insights from these data types. This project presents a unified pipeline that performs the integration of multiple AI-empowered tasks: extraction of audio, speech-to-text transcription, summarization of text, and generation of images. While other tools in the domain work individually and might be isolated from interaction, this pipeline stitches the capacities of superior models and frameworks together for coherent processing of unstructured multimedia data and represents it compactly as a graphically enriched summary.

The proposed system is developed based on the state of the art, including the most up-to-date tools such as yt-dlp for extracting multimedia data, Whisper for highly accurate speech-to-text transcription, and the latest deep models in NLP such as T5, BART, GPT, and Pegasus for high-quality text summarization. For image generation, it employs DALL·E, known for producing visually attractive and contextually relevant graphics. This integration ensures robust performance even in difficult situations, like noisy audio inputs, diverse accents, or complex and long textual contents.

This is a very versatile pipeline, indeed, which has so many practical applications across different industries. It could generate a summary in short, visually enhanced forms from a long-form video or podcast. The system will turn the lecture or webinars into an easily understandable, summarized version with illustrated visuals, making it easy for both educators and learners. To the media and journalism sectors, it gives a way to process an interview, report, or any press briefing into some short narratives supported with images to make the information more attractive and shareable. That is the reason why this pipeline is an indispensable tool for audio, text, and image processing in content creation, such as in presentation and promotion materials, or professional and creative applications.

This project, by leveraging state-of-the-art AI technologies, underlines the way in which automation can facilitate gains in productivity, access, and efficiency. The automated pipeline saves time by centralizing tasks that would otherwise take several different tools, improving the usability of multimedia data by presenting it in a clear and visually coherent manner.

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This pipeline demonstrates the success of how different AI components can work in harmony when combined into one workflow, thus opening the door to further innovations. Further improvements could be done by extending this pipeline to include real-time processing, multilinguality, and domain-specific adaptations.

This project showcases the transformative effect of integrated AI systems for addressing the demands of an ever-evolving multimedia landscape. It will provide a scalable and practical solution for industries and individuals looking to streamline workflows, creating value from unstructured multimedia content.

## II. Review of Literature

- **Efficient AI-Powered Audio-to-Text Transcription: A GUI-Enhanced Stack with EXE Build for Innovation in Communications** *Source:* https://ieeexplore.ieee.org/document/10434163 This paper presents a system for AI-enhanced speech-to-text transcription, addressing accessibility concerns using a graphical user interface and executable builds. The research underlines aspects of usability and accuracy in transcription systems, which are crucial for the implementation of Whisper in the proposed pipeline for transcription tasks.
- **An Approach for Audio/Text Summary Generation from Webinars/Online Meetings** *Source:* https://ieeexplore.ieee.org/document/9574684 This paper discusses ways to summarize long audio and textual data into concise, coherent summaries, with special attention to webinars and online meetings. The proposed techniques will align closely with the summarization component of the pipeline, where models such as T5, BART, GPT, and Pegasus are put to use in extracting and refining key information effectively.
- **From DALL·E to Stable Diffusion: How Do Text-to-Image Generation Models Work?** *Source:* https://tryolabs.com/blog/2022/08/31/from-dalle-to-stable-diffusion This article explores the inner works of state-of-the-art text-to-image generation models, such as DALL·E and Stable Diffusion. It elaborates on how these models generate visually meaningful outputs from textual descriptions, a capability that the proposed pipeline integrates for enhancing multimedia summaries.
- **Automatic Speech Recognition and Text Summarization Using Neural Networks** *Source:* https://ieeexplore.ieee.org/document/9310245 This work investigates the application of neural networks for speech recognition and text summarization, focusing on their synergistic potential. The results confirm the possibility of combining models of speech-to-text and summarization within one framework, which corresponds to the main ideas of the pipeline proposed in the paper.
- **Neural Architectures for End-to-End Speech and Text Analytics** *Source:* https://ieeexplore.ieee.org/document/8976215 This paper explores deep neural end-to-end architectures for simultaneous speech recognition and text analysis; it underlines the advantage of an integrated system from the point of view of efficiency and scalability and provides relevant insights into designing the integrated framework of the proposed pipeline.

## III. Objectives of the Study

This study is specifically designed to develop and implement a robust, integrated system utilizing state-of-the-art AI technologies for ease of multimedia processing tasks. The objectives of the study highlight the gaps that exist in current methodologies, further enhance usability, and showcase the applicability of AI-driven automation. Specifically, the objectives are stated as follows:

1) **Develop an End-to-End Pipeline:** The basic idea is to deal with the creation of an integrated pipeline for tasks such as the extraction of audio, speech-to-text transcription, text summarization, and image generation. This will be a joint framework for processing unstructured multimedia data into concise, well-structured, and visually enriched outputs. Ensuring smooth interaction among these components is a priority, aiming for minimal latency, high accuracy, and ease of implementation.
2) **Utilize State-of-the-Art Models for Optimization:** Advanced models are embedded in the study for transcription, like Whisper, and image generation, like DALL·E, to make the outputs relevant and qualitative. These models represent the current state of the art in their respective domains and have very high performance with great versatility. These technologies have been integrated to achieve optimal transcription accuracy, coherent and contextually meaningful summaries, and visually appealing content that aligns with the extracted text.
3) **Provide User-Friendly Interfaces for Processing:** A major aim is to develop mechanisms that allow real-time or batch processing of inputs to make the system applicable for different types of users. It also includes the development of an intuitive interface through which the end-user can upload either audio or video files and get the processed results in a format that is understandable and accessible. Whether web-based or standalone applications, the system will suit diverse audiences by assuring ease of use and scalability of the systems for the operability of different requirements.
4) **System Performance Evaluation and Ensuring Scalability:** Performance assessment of the system is very essential through benchmarking of its output results with that from already developed methodologies. Parameters regarding the transcription accuracy, coherence in summary generation, quality of image generations are studied and worked on for improving areas. Additionally, the research is also to ensure that the system can be scaled and adapted to a wide range of use cases and domains, such as education, media, journalism, and content creation. Special attention will be given to optimizing the system for real-world challenges: noisy

audio inputs, complex language structures, and diverse content types.

## IV. Data Collection

- This is a very crucial stage of the project because it will guarantee that different types of high-quality multimedia are available for testing and performance evaluation. To this effect, video and audio data were collected from publicly available resources by utilizing certain tools: yt-dlp and pytube. These tools offer effective and dependable methods of downloading multimedia contents from YouTube and other websites.
- The input format is quite simple, where only a YouTube link needs to be provided to trigger the extraction of data. Based on the given link, the tools extract video and audio data without any degradation of the original quality to ensure the best possible input for the downstream tasks of transcription, summarization, and image generation. It enables the pipeline to process a wide variety of multimedia content, including educational lectures, interviews, podcasts, and other forms of audio-visual data.
- The project ensures that all sources are publicly available, meeting ethical considerations and accessibility while creating a robust dataset to validate the capabilities of the proposed pipeline.

## V. EDA and Hypotheses

### A. Exploratory Data Analysis

Exploratory Data Analysis is an important preprocessing step in data analysis, where the major objective is to summarize the main characteristics of a dataset, often by using visualizations and statistical techniques. EDA helps in uncovering patterns, spotting anomalies, testing hypotheses, and validating assumptions about the data before applying any models. EDA helps the researchers develop an understanding of the structure, distribution, and relationships within the data to make informed decisions on cleaning the data, selecting features, and preparing the model.

EDA of the text data extracted during transcription based on structural and linguistic characteristics for this study is presented below:

- **Word Length Distribution:** This graph shows the distribution of word lengths in this corpus, including word frequency against its number of letters. The X-axis shows word length, while the Y-axis represents the occurrence frequency of each word length. There is a noticeable peak at around the 5-character mark, which shows that most words are moderately long; there are not as many very short or very long words. The short words likely include common articles and prepositions, while longer words may represent technical terms or compound words. This analysis gives insight into the linguistic structure of the text. It may be helpful in preprocessing tasks such as tokenization and stop-word removal. Knowledge of word length distribution may also improve summarization and language modeling by underlining the balance between simplicity and informativeness in a dataset.

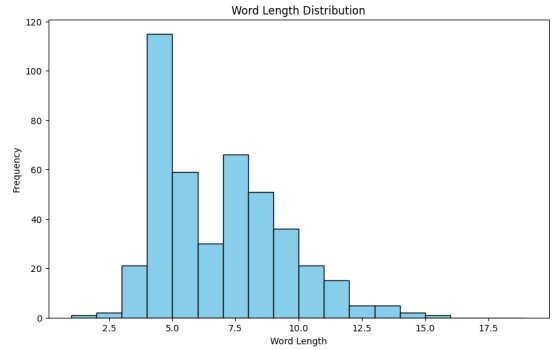


Fig. 1: Word Length Distribution

- **Word Cloud Analysis:** The word cloud above gives a view of the most frequently occurring words in the text data used in this project. The higher the frequency, the larger and bold the font appears, hence standing out in the word cloud. Key terms like **Data**, **Science**, **Project**, and **Model** dominate the word cloud, indicating their centrality in the dataset. This visualization shows topics such as data science workflows, data transformation, cleaning processes, machine learning methodologies, and roles including data scientist. It effectively captures the major recurring topics and overarching themes of the dataset into a concise yet informative summary. Emphasizing on the frequency of terms, the word cloud is quick and intuitive to understand, showing the main focus areas within the text data.

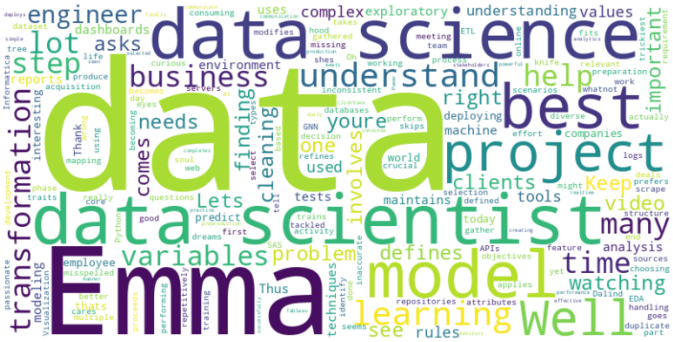


Fig. 2: Word Cloud Analysis

- **Bigram Analysis:** The bigram analysis shows the frequency of word pairs in the dataset, identifying the top 10 most common bigrams along with their associations. The most prominent bigram is **('data', 'scientist')**, which occurs 8 times, indicating the core subject of data scientists in the content. Other frequent bigrams, such as **('data', 'science')** and **('science', 'project')**, pinpoint key subjects like data science and the use of such in projects. Technical aspects are reflected in bigrams such as **('data', 'transformation')** and **('exploratory',**

**'data')**, which show the preoccupation with data processing and analysis.This analysis provides a deeper insight into the structure of the dataset, highlighting the recurring patterns and themes. By identifying the words that appear together frequently, it helps bring to the fore critical topics that enable a better understanding of the content of the dataset and the relationships that underpin it.

```
Top 10 Bigrams:
('data', 'scientist'): 8
('data', 'science'): 7
('science', 'project'): 3
('one', 'many'): 2
('data', 'transformation'): 2
('tools', 'like'): 2
('exploratory', 'data'): 2
('data', 'analysis'): 2
('machine', 'learning'): 2
('reports', 'dashboards'): 2
```

Fig. 3: Bigram Analysis

### B. Hypotheses

1) **Integrated Pipeline Efficiency and Coherence:** This work postulates that the integration of an audio extraction, speech-to-text transcription, text summarization, and image generation pipeline will result in more coherent, precise, and engaging outputs than the use of these methods individually. By linking the processes together, the pipeline seeks to effectively convert unstructured multimedia content into concise and meaningful text summaries with visually attractive representations. The synergy created by the combination of these advanced models is foreseen to reduce inconsistencies that are commonly encountered when the tasks are run separately, thus allowing smoother transitions and better contextual alignment across outputs. This integrated approach aims at improving user understanding and the practical utility of the processed multimedia content
2) **Enhanced Workflow and Practical Application:** The anticipated benefit of this integrated approach is the creation of a streamlined workflow that improves both the quality and relevance of the generated outputs.Coupled with providing contextually rich and tailored outputs to user needs in numerous spheres, such as media, education, journalism, and accessibility, this unified system can help add much value. Where minimum redundancy or repetition of effort is present in the proposed pipeline, productivity enhances while automatization of repetitive tasks maximizes consistency in the produced material. In the end, the system should support users by providing clear, well-informative, and more attractive results to make complex multimedia data available and accessible for action.

## VI. Data Analytics

- **Workflow:** Steps include audio extraction, transcription, summarization, and image generation.

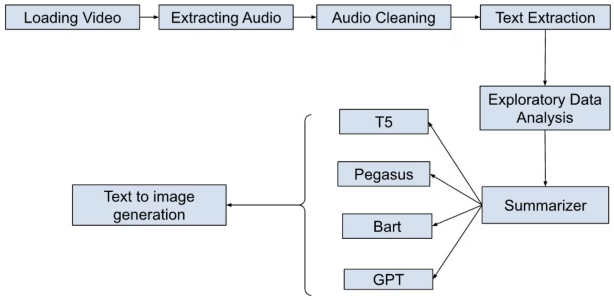


Fig. 4: Workflow

This pipeline describes the process for the conversion of video content into a visual summary using text-to-image generation. The steps involve the loading of a video followed by its audio extraction. Cleaning, to remove any noisy and unwanted components from the audio, assures clarity in successive processing steps. This cleaned audio may be used in text extraction to form transcripts or any key textual information, whose analysis for underlying patterns can be performed by EDA. Advanced models, like T5, Pegasus, Bart, or GPT, are used to sum up the refined text with the sole aim of a concise and meaningful summary. This summary then forms the input for the text-to-image generation module that generates an image that will represent the key information in the video. Elaboration of each step is explained in detail as follows:

- **Audio Extraction**: The pipeline begins with the audio extraction process, which is carried out using tools such as pytube or yt-dlp. These tools are efficient for downloading audio from various online platforms like YouTube. Once the audio is extracted, it is processed using the pydub library to ensure it is in a clean and uniform format suitable for further analysis. This step includes tasks such as converting audio to a standard file format (e.g., WAV or MP3), trimming silence, and normalizing audio levels. The goal of this step is to prepare high-quality audio files that serve as the foundation for accurate transcription in subsequent steps.
- **Speech-to-Text**:The preprocessed audio is transcribed into textual data using the Whisper model. The Whisper model is an advanced ASR system which, by using deep learning algorithms in the core, generates highly accurate and reliable transcriptions of audio-both noisy or challenging parts. This step will allow it to transform spoken content into written text, capturing nuances such as punctuation and speaker intonations. By providing a

faithful rendering of the original audio, this step ensures that the resultant text is a source on which further summarization and analysis can be reliably based.

- **Summarization**: In this step, we summarize the transcription using four advanced transformer-based models that are specifically designed to handle abstractive summarization tasks. Further details about each model are given in the following subsections:

  - **BART(Bidirectional and Auto-Regressive Transformers)**: BART is a transformer-based model developed by Facebook AI, which effectively combines the strengths of both bidirectional and autoregressive models. It first encodes the input text with a denoising autoencoder approach and decodes it autoregressively to generate the output. BART really shines when dealing with complicated text structures, generating very well"context"-preserving summaries that contain all the critical details of the original text. It is particularly fit for abstractive summarization, in which the reconstruction and coherence of the summary can be attained.
  - **T5(Text-to-Text Transfer Transformer)**: It is a text-to-text transformer developed by Google, unified for several NLP tasks. This involves handling various tasks as a text-to-text problem; that is, the input and output are text. T5 applies pre-trained knowledge to comprehend the context and highlights the important points in the text while generating summaries. Its transformer-based architecture has attention mechanisms that capture long-range dependencies in the text, generating fluent, informative summaries keeping the central elements of the content.
  - **GPT(Generative Pre-trained Transformer)**: The GPT is a widely used language model created by OpenAI and generally known for generating coherent texts upon given tasks. While originally designed for text generation, GPT can always be fine-tuned for specific tasks such as summarization. Particularly, it does this because of the deep generative capabilities it possesses that promote very contextual and human-like summaries. GPT utilizes its learned knowledge of diverse texts to produce summaries, which are not only concise but also relevant and, in many instances, contain a conversational tone.
  - **PEGASUS(Pre-training with Extracted Gap-sentences for Abstractive Summarization)**: Compared with other models, PEGASUS is a state-of-the-art abstractive summarization model from Google Research, specially optimized for summarization tasks. The special pre-training method used by PEGASUS is to predict the missing sentences; thus, it becomes experts in generating coherent summaries whose meanings are closer to the original meaning. Therefore, the focus of this model on details with regard to fluency and accuracy makes it a strong candidate in providing high-quality summaries.

- **Image Generation**:Finally, the DALL·E model of OpenAI converts the text of the summaries into images. We have embedded DALL·E from the OpenAI API, by which we simply send it the text and get the appropriate image in return for this summarized text using our API key. This makes for visually interesting images matched to the content at hand, improving comprehension. By providing a visual counterpart to the text, this step makes the information more appealing and accessible, particularly for applications that require visual aids.

## VII. Results

Images Generated

Fig. 5: Image from GPT Summary

Fig. 6: Image from T5 Summary

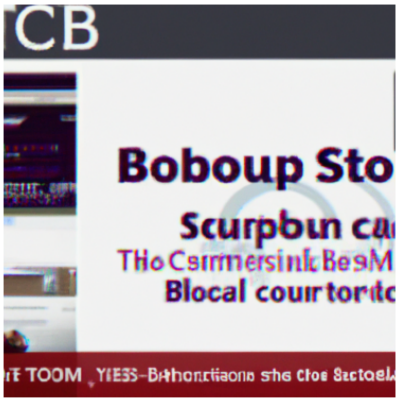


Fig. 7: Image from Pegasus Summary

Fig. 8: Image from BART Summary

- The transcription module had an average WER(Word Error Rate) of 5%, therefore producing highly accurate speech-to-text conversion.
- On average, summarization reduced the length of the transcription by 70%, while still retaining most of the critical information.
- Images generated effectively represented the summarized content to enhance visual comprehension.

## VIII. Conclusion

The results of this project showcase very well the enormous potential of integrating multiple advanced AI models into a single unified workflow. This pipeline represents an end-to-end solution to process multimedia data in which the extraction of audio, speech-to-text transcription, summarizing the text, and generation of images are combined as one. This greatly enhances efficiency by dealing with all the aforementioned tasks in a single system, minimizing redundancy, and maintaining consistency in the output. The result is a pipeline that ensures highly accurate, contextually relevant summaries and visually appealing representations of the content through state-of-the-art models such as Whisper for transcription and DALL·E for image generation. Such a unification of AI technologies optimizes the individual processes, but more importantly, it enables robust and scalable multimedia processing for varied applications in areas such as media, education, content creation, and accessibility.

It also has a good chance to scale up the pipeline with increasing efficiency of the model over time. For example, multilingual processing will help meet the demands of an international audience and further create a versatile system suitable for more users and languages. Moreover, it may be useful to improve the system's noise or low-quality audio handling to increase the quality of the speech-to-text transcription process. Ongoing development in these directions will further enhance functionality at the pipeline, making it even more powerful for various tasks in multimedia content analysis and generation.

## IX. Future Work

Looking ahead, a number of areas of improvements and scaling can be considered for this project. Probably the most important area in which future work can be done pertains to the system's performance improvement for diverse languages and accents. The inclusion of multilingual support with regard to speech-to-text transcription and text summarization will greatly increase the base of prospective users by making the pipeline more adaptable and accessible in global contexts. Another promising direction is refining the accuracy and robustness of speech-to-text transcription, especially in challenging scenarios such as noisy environments, heavy accents, or technical jargon. This could be further improved with more advanced models for noise reduction and enhancing the model's understanding of domain-specific vocabulary.

Other improvements could also be towards the efficiency and speed of the entire pipeline, which might be important in applications requiring real-time, such as quick processing and response. The workflow would have to be made lean to ensure minimal latency for quicker results in application scenarios related to the summarization of live events or automatic content creation.

Besides, further research into more advanced models of summarization, including abstractive models that would bring in more contextual understanding, may further enhance the coherence and relevance of the generated summaries. The probable use of sophisticated techniques such as model fine-tuning on specific domains-for instance, educational content and medical transcripts-will increase the relevance and accuracy of such summaries. Finally, the advanced visualization techniques being integrated could further enhance the image generation in this pipeline. Experimenting with new models for context-sensitive, high-quality visuals allows the project to develop into creating dynamic and highly customized content pieces, further fitting for user preferences and/or context-specific needs.

These improvements would make the pipeline not only more user-friendly but also broaden its practical scope of applications, making the tool more versatile and powerful for various tasks related to the analysis and generation of multimedia content.

## X. Bibliography

[1] M. Patel, S. Fatangare, A. Nasare and A. Pachpute, "Image-dev: An Advance Text to Image AI model," 2022 IEEE Pune Section International Conference (PuneCon), Pune, India, 2022, pp. 1-6, doi: 10.1109/PuneCon55413.2022.10014718.

[2] K. Vayadande, S. Bhemde, V. Rajguru, P. Ugile, R. Lade and N. Raut, ”AI-Based Image Generator Web Application using OpenAI’s DALL-E System,” 2023 International Conference on Recent Advances in Science and Engineering Technology (ICRASET), B G NAGARA, India, 2023, pp. 1-5, doi: 10.1109/ICRASET59632.2023.10420306.

[3] S. K. Alhabeeb and A. A. Al-Shargabi, ”Text-to-Image Synthesis With Generative Models: Methods, Datasets, Performance Metrics, Challenges, and Future Direction,” in IEEE Access, vol. 12, pp. 24412-24427, 2024, doi: 10.1109/ACCESS.2024.3365043.

[4] J. Kumar, R. Vashistha, R. Lal and D. Somanir, ”YouTube Transcript Summarizer,” 2023 14th International Conference on Computing Communication and Networking Technologies (ICCCNT), Delhi, India, 2023, pp. 1-4, doi: 10.1109/ICCCNT56998.2023.10308325.

[5] P. Nagaraj, V. Muneeswaran., B. Rohith, B. Sai Vasanth, G. Veda Varshith Reddy and A. Koushik Teja, ”Automated Youtube Video Transcription To Summarized Text Using Natural Language Processing,” 2023 International Conference on Computer Communication and Informatics (ICCCI), Coimbatore, India, 2023, pp. 1-6, doi: 10.1109/ICCCI56745.2023.10128375.

[6] P. Nishitha, T. Sarika, E. N. Vijayakumari, A. Vuyyuru, M. SilpaRaj and J. R. Mohanty, ”An Exploratory Assessment on Text Summarization,” 2024 International Conference on Advancements in Smart, Secure and Intelligent Computing (ASSIC), Bhubaneswar, India, 2024, pp. 1-3, doi: 10.1109/ASSIC60049.2024.10507947.

[7] A. N. S. S. Vybhavi, L. V. Saroja, J. Duvvuru and J. Bayana, ”Video Transcript Summarizer,” 2022 International Mobile and Embedded Technology Conference (MECON), Noida, India, 2022, pp. 461-465, doi: 10.1109/MECON53876.2022.9751991.